

Hema M

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Summary

Artificial Intelligence and Data Science graduate with practical experience in AI, Machine Learning, and Software Development. Proficient in Python and modern development tools, with experience gained through academic and personal projects. Committed to continuous learning and eager to contribute technical skills while growing in a professional environment.

Internship

MARKDOT PVT LTD, AI & ML Intern – Onsite, India

Dec 2024 – Feb 2025

- Developed AI/ML solutions using NLP, Sentence-BERT, and YOLOv5 for text processing and object detection applications.
- Conducted data preprocessing, model training, and performance evaluation to support project requirements.
- Collaborated on model implementation and validation across machine learning workflows.

SYNCNER, ML Engineer Intern (Learning Exposure) – Remote, India

Jun 2025 – Dec 2025

- Built machine learning solutions using Python and CNN architectures for classification and prediction tasks.
- Integrated APIs, including Twilio, to support automated communication workflows within applications.
- Assisted in model optimization, testing, and deployment activities to enhance performance and reliability.
- Worked in a collaborative remote environment, following industry development and project management practices.

Projects

Accident Detection and Alert System

- Developed a real-time accident detection system using YOLOv5 and OpenCV to identify accidents from video streams.
- Integrated Twilio API and Flask to automate emergency notifications and system deployment.
- Gained hands-on experience in computer vision, API integration, and real-time alerting workflows.

AI-Based Banking Chatbot

- Built an NLP-powered chatbot using Sentence-BERT to understand and respond to banking-related queries.
- Implemented semantic search with precomputed embeddings and Flask REST APIs for efficient query processing.
- Improved response accuracy by leveraging embedding-based similarity matching techniques.

AI-Powered Healthcare Assistant

- Developed a healthcare assistant for disease prediction using TF-IDF and Random Forest models based on user symptoms.
- Applied CNN architectures including ResNet and VGG16 for medical image classification tasks.
- Built and integrated machine learning models using Flask, Scikit-learn, and TensorFlow/Keras for end-to-end prediction workflows.

Skills

Programming Languages:	Python, JavaScript
Frontend:	HTML/CSS, ReactJS
Databases:	MongoDB, MySQL
Libraries & Frameworks:	Fast Api, React.js, Flask, Scikit-learn, Pandas, Matplotlib
Other Tools:	Git, GitHub, Postman API, Vercel
Soft skills:	Adaptability, Communication, Problem Solving, Team Collaboration

Certifications

1. Microsoft Certified – Azure AI Fundamental
2. IBM – Enterprise Design Thinking
3. Databricks – Generative AI fundamentals
4. Nasscom – Digital Application Fundamentals (STEM)

Education

SNS College Of Engineering -B.Tech AI & DS - CGPA : 8.29/10

2022 – 2026